Mead App Guide – v1.0

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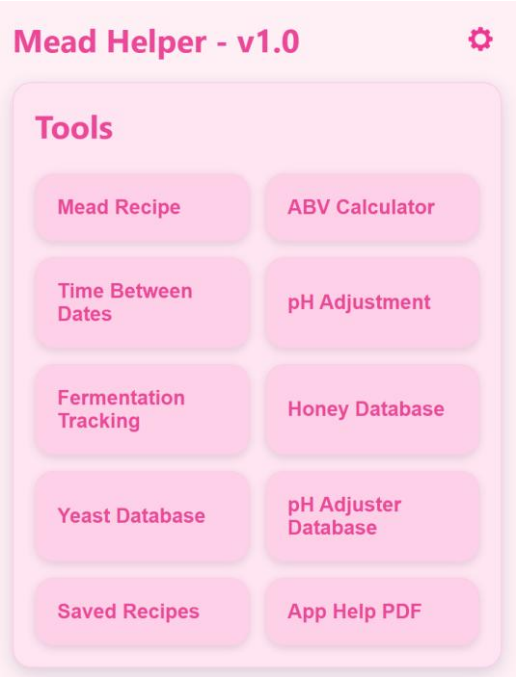
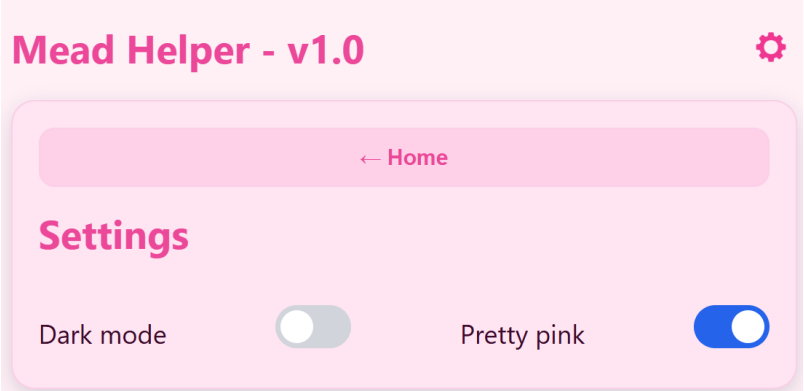
Front Screen

When you open the app, you are taken to the home screen as shown below. From here there is everything that you need to help design your mead. To find out more about each button then go to the Table of Contents at the start of this document and find what page to go to for each section.



Settings

In the top right corner of the home screen is a gear icon to indicate settings. When pressed it takes you to the following screen:

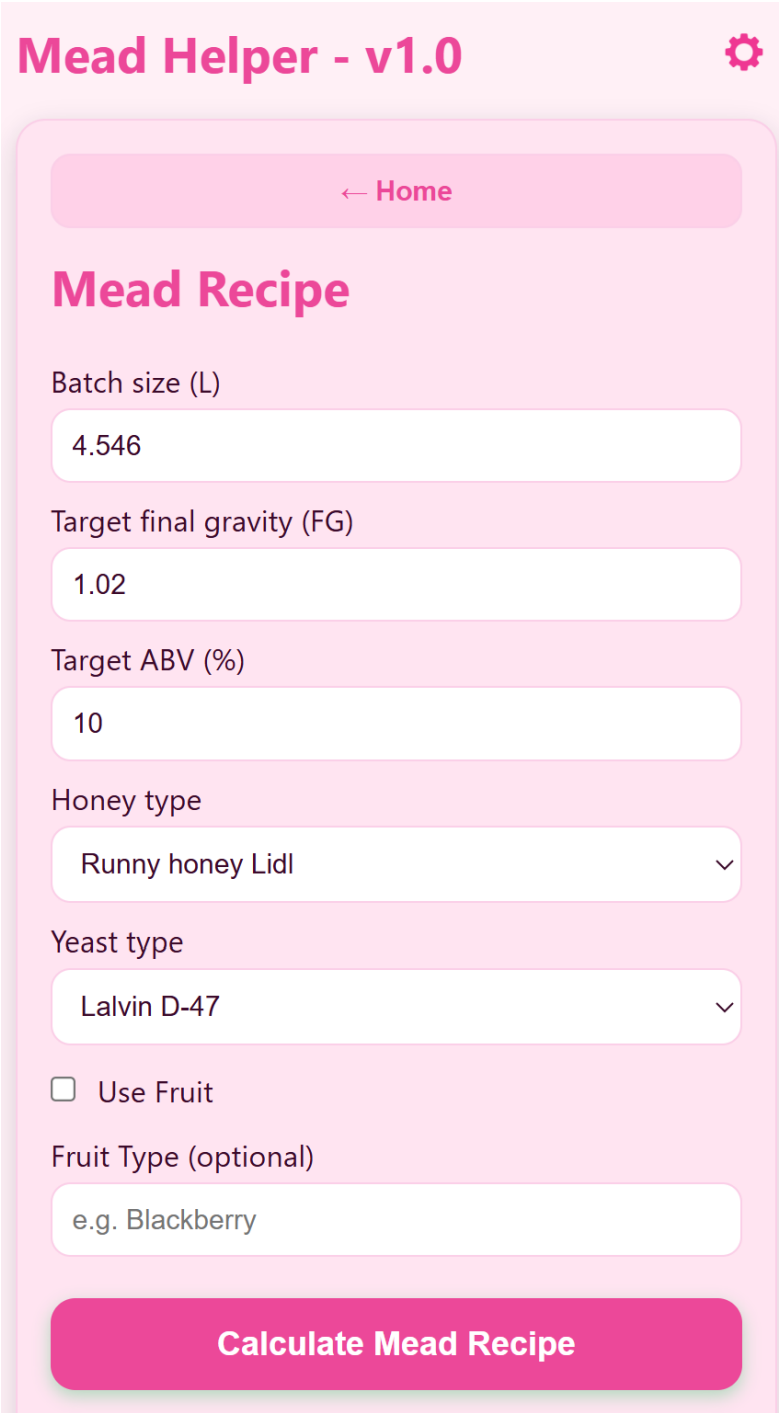


This lets you choose between 3 colour palette options:

1. Pretty pink
2. Dark mode
3. Light mode

To toggle to “Pretty pink” and “Dark mode” you just need to click the slider to the right of them. For Light mode you instead need to click the slider that is currently engaged, for the image above you would click the Pretty pink slider.

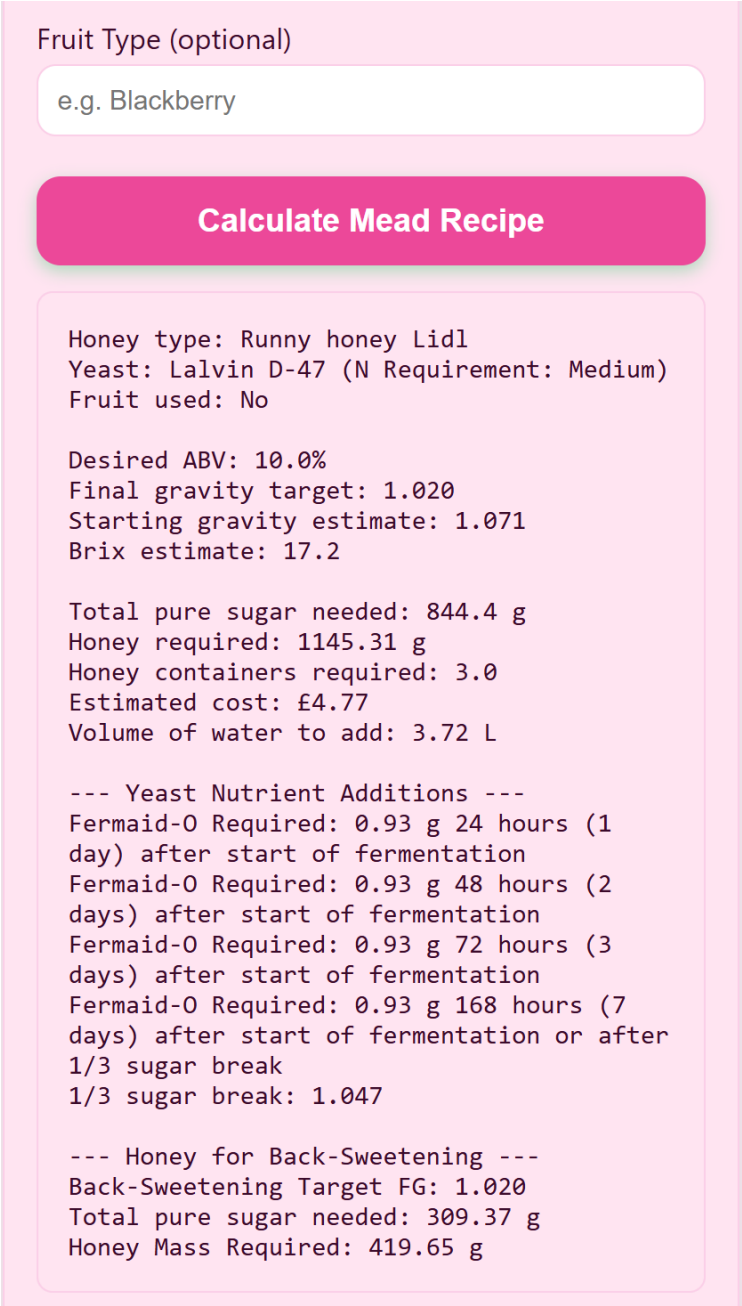
Mead Recipe



This screen is used for designing a mead based on the following 5 parameters:

1. Volume of mead desired
2. The desired final sweetness level for the final mead - a final gravity of 0.999-1.006 is considered dry, 1.006-1.015 is considered a medium mead, 1.012-1.02 is considered sweet and 1.02+ is a dessert mead
3. The desired ABV of the mead - yeast cannot survive/produce alcohol with an ABV above 21% on their own, to achieve these high ABVs extra supplements are required
4. The honey being used as the sugar source for the mead
5. The yeast being used to convert the sugar in the honey into alcohol

There is a “Use Fruit” checkbox and an entry field for the fruit being used, however the app is still under development, and these currently do nothing towards the mead recipe. The values currently in place are for a basic mead and allow for anyone who isn’t too into homebrewing to have a nice mead to start with.



Once the “Calculate Mead Recipe” button has been clicked it will bring up the following data, allowing you to brew your own mead to the specifications that were outlined.

At the very bottom of this screen there are 2 more buttons:

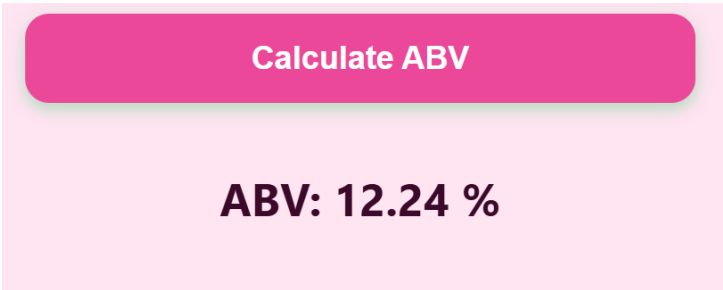
The “Save Recipe” button takes the outputted data above and stores it under the “Saved Recipes” button. The “Export as PDF” button takes the outputted data above and copies it into a PDF where you can save it locally to your device or the cloud, but also to print the data, allowing you to have a physical copy of your recipe.

ABV Calculator

Once a specific gravity has been reached in the fermentation process an ABV check might be desired. This is where the “ABV Calculator” button comes in. Once clicked it will take you to the following screen:

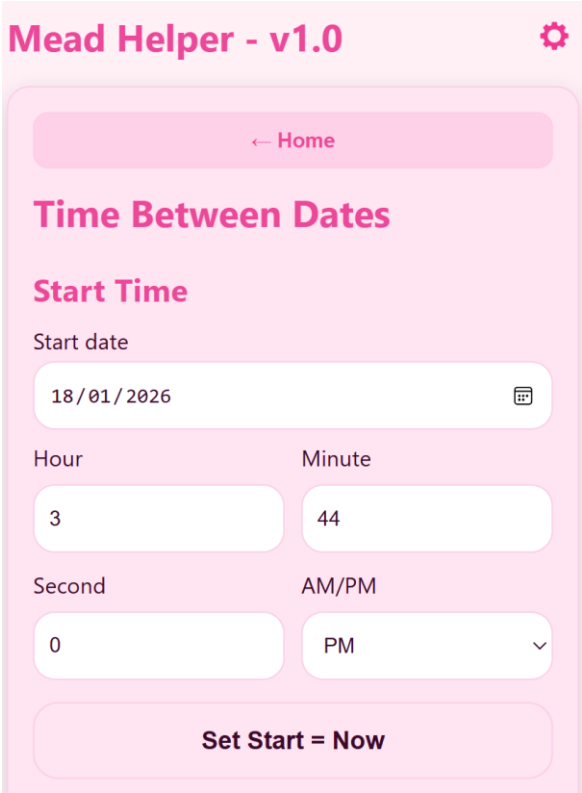
The only values required are the Starting Gravity and the Final Gravity. The Starting Gravity is the specific gravity measured at the start of the brew before the yeast was added, the Final Gravity is the specific gravity reading taken during the fermentation process where an estimated ABV is desired to be known.

Once the “Calculate ABV” button is pressed it calculates the ABV from the values entered, the example output is shown below:

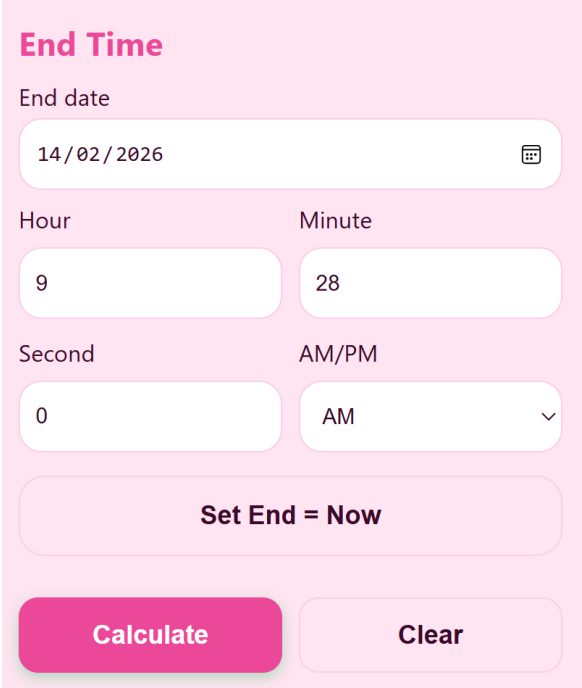


Time Between Dates

This screen calculates the decimal value between two known dates. This is an especially important value to know as this helps track fermentation. Once the “Time Between Dates” button is pressed the following screen comes up:

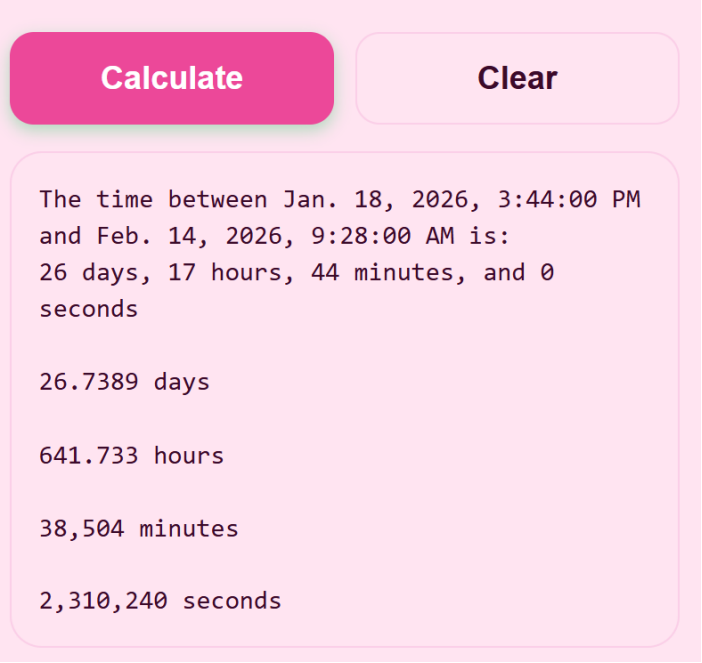


This is the earlier date that is to be entered, the “Set Start = Now” auto inputs the current time into the entry fields.



Scrolling down brings the second time entry field; it works the same way as the “Start Time” except this is the date that is further along chronologically. The “Calculate” button runs the program, the “Clear” button voids the output to ensure that the data stated is the one that is desired.

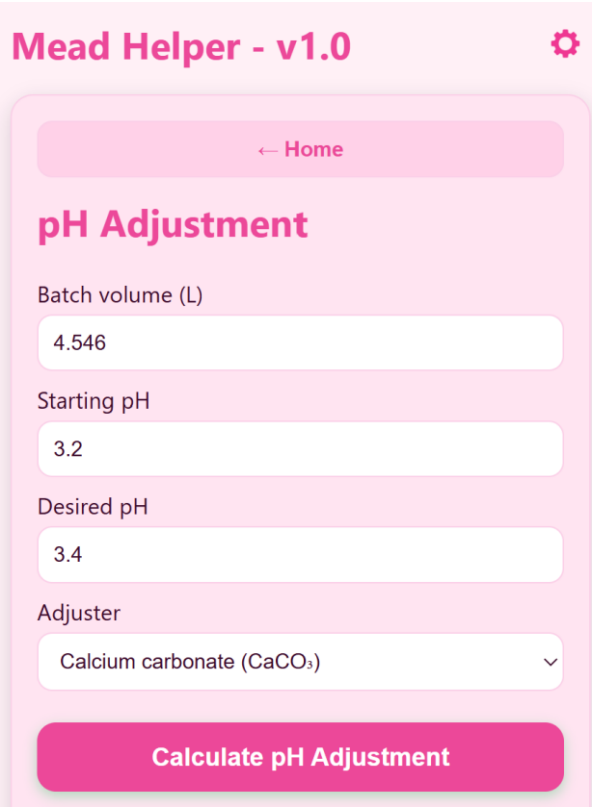
Once the “Calculate” button is pressed the calculator finds the time difference and displays it as so:



This is the output from the example entry fields. For the fermentation tracking page the “26.7389” value is to be copied over as the fermentation tracking page runs time in the unit of “days”.

pH Adjustment

When homebrewing the pH might be unfavourable for the yeast to grow in. To ensure accurate adjustments the “pH Adjustment” page was created. Once the “pH Adjustment" button is pressed the following screen comes up:

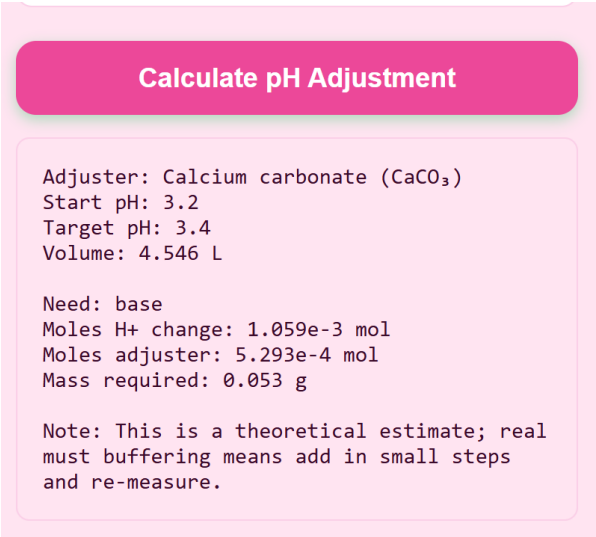


There are 4 parameters that need to be specified:

1. The volume of mead
2. The starting pH
3. The desired final pH
4. The chemical pH adjuster being added

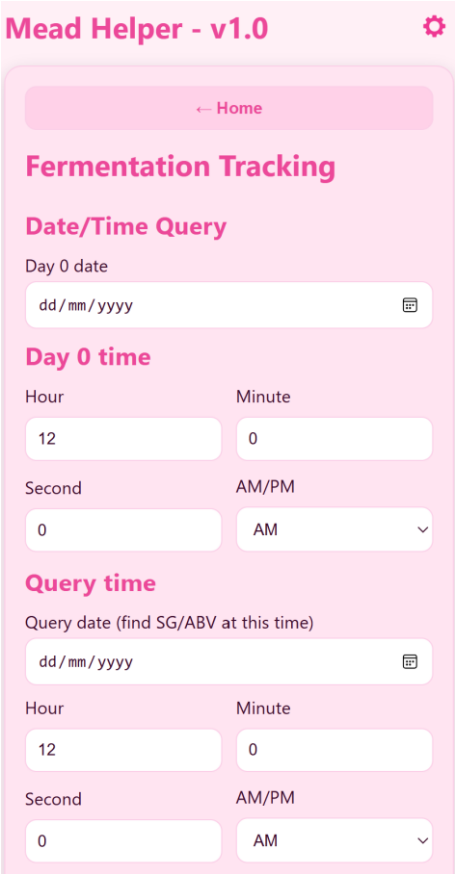
It is important to note that if the pH is to be reduced, the chemical pH adjuster being used should be an acid. If the pH is to be raised, the chemical pH adjuster being used should be a base.

The example values produce the following output:



Fermentation Tracking

This screen is the most powerful when it comes to homebrewing, as it allows you to plan ahead and tell if your mead is progressing well. Once the “Fermentation Tracking” button is clicked the following screen will come up:



It is built much like the “Time Between Dates” pages, the Day 0 date and time is the date and time at which the yeast is inoculated with the yeast and honey. The next set of inputs required are the following:

Starting SG

1.071

2nd SG

1.06

Time for 2nd SG reading

12.8993

3rd SG (if no 3rd point then leave blank)

Time for 3rd SG reading (if no 3rd point then leave blank)

Initial mass of yeast (g)

5

Volume of brew (L)

4.546

Graph until day

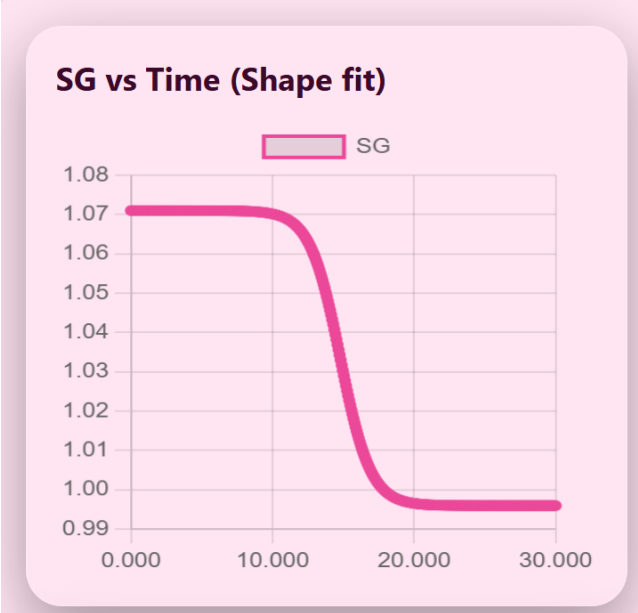
30

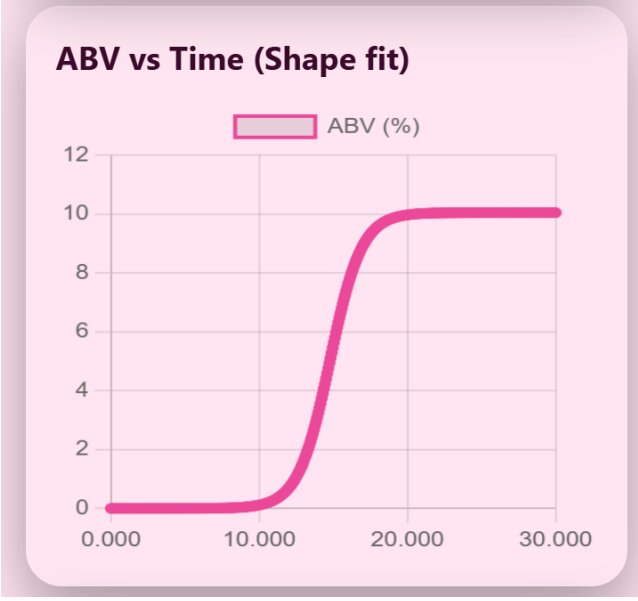
Calculate Fermentation

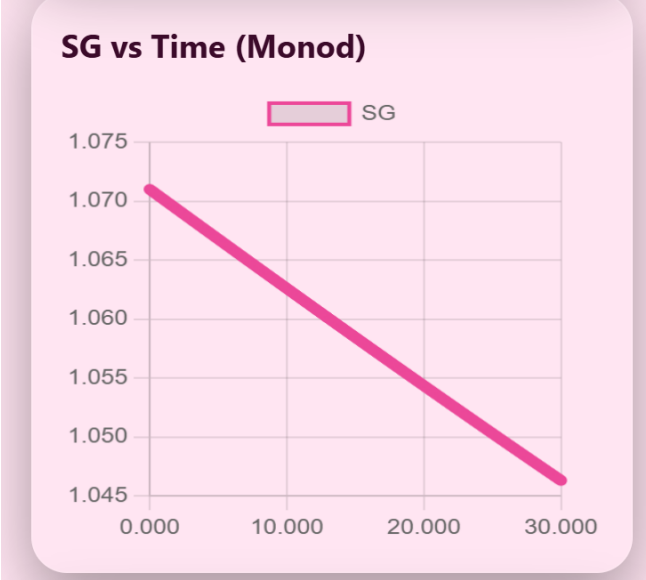
- The “Starting SG” is the specific gravity reading taken at the time just before the yeast was inoculated
- The “2nd SG” entry is the first specific gravity reading taken during the fermentation process
- The “Time for 2nd SG reading” entry is the time of the second reading taken in units of days (hence the requirement for a “Time Between Dates” page)
- The “3rd SG” and “Time for 3rd SG reading” entries are the same as the 2nd, but for a reading taken further along
- The “Initial mass of yeast” entry comes from what is written on the pack of yeast used to inoculate the mead
- “Volume of brew” entry is the amount of mead being made
- “Graph until day” entry is how far along you want the graph to go, this may need to be increased if the fermentation is progressing slowly

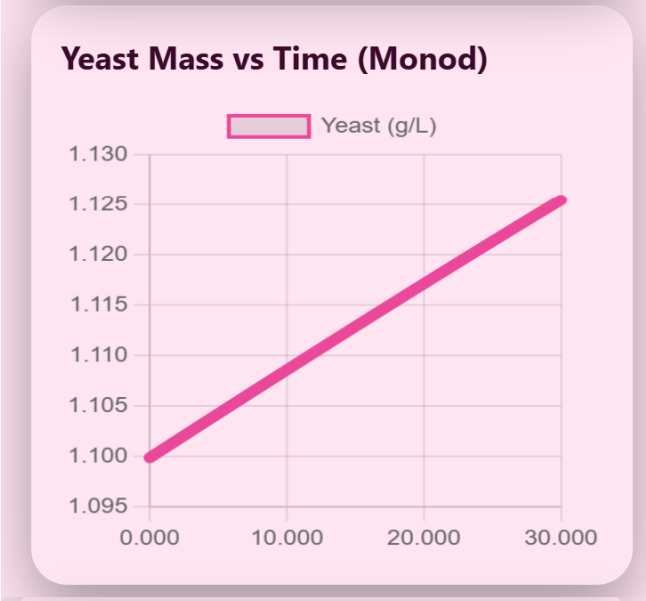
There are 5 graphs that are produced when the “Calculate Fermentation” button is clicked:

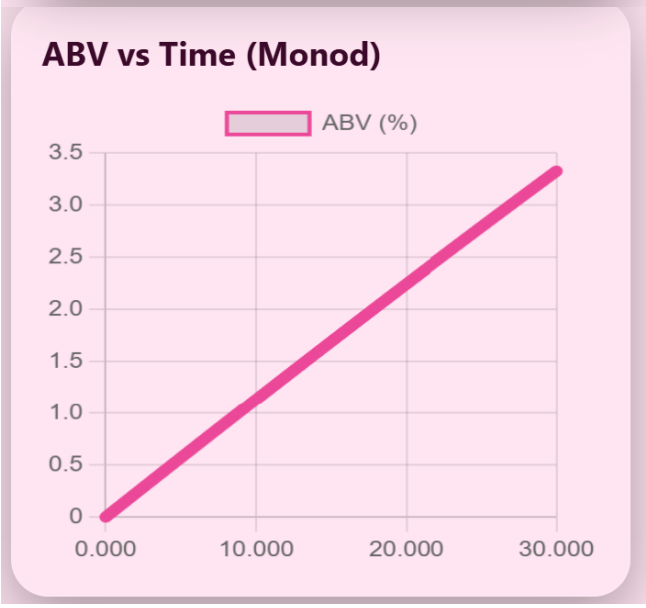
- SG vs Time (Shape fit)
- ABV vs Time (Shape fit)
- SG vs Time (Monod)
- Yeast Mass vs Time (Monod)
- ABV vs Time (Monod)



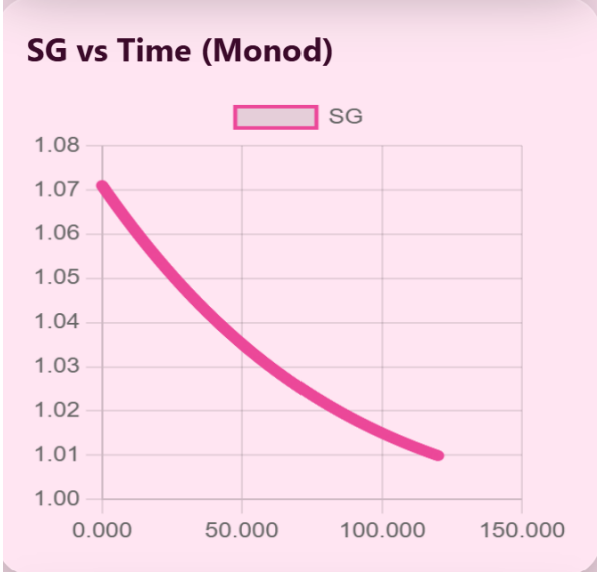




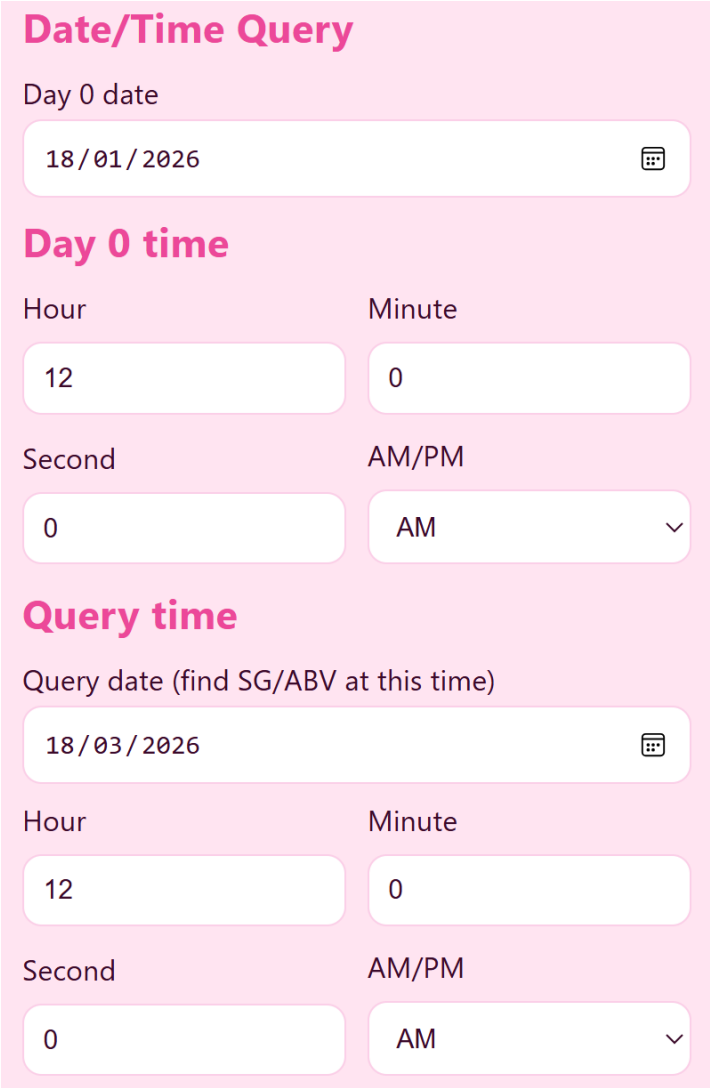




These graphs are a perfect example of when you might want to increase the “Graph until Day” entry, as when that value is changed to “120” the “SG vs Time (Monod)” graph becomes:



To find the specific gravity value at a specific time the initial entry fields are used. Keeping the rest of the inputs the same and adding the following dates/times:



Gives the following predictions for the fermentation:

```
Time between Day 0 and Query is:
59 days, 0 hours, 0 minutes, and 0 seconds

59.0000 days
1416.000 hours
84,960 minutes
5,097,600 seconds

--- Predictions at t = 59.0000 days ---
Shape fit: SG = 0.9960, ABV = 10.04%
Monod:     SG = 1.0306, ABV = 5.43%
Monod yeast concentration: 1.1417 g/L
```

For the specific brew that this data came from the Monod values match reality much more closely.

Honey Database

This screen holds the data for the honeys you can use in your recipes, it holds data on its sugar content and price so it can help you plan out your recipe on a financial level. When the page is opened this appears:

Honey name	Sugar (%)	Price per bottle (£)	Mass per bottle (g)	
Runny honey Lidl	79.7	1.59	400	×
Clear honey Lidl	79.7	1.69	500	×
Generic honey	80	1.49	300	×
Runny honey Rowse	80.8	5	720	×
Manuka honey Lidl	85.9	4.89	250	×

This is all the core data that the app needs to carry out the calculations. All this data can come from the label of the honey you are using except the price per bottle which can be read in the shop.

To add new honeys into the database is easy, as below this main data section there is an entry field for a new honey:

Add new honey

Orange Blossom

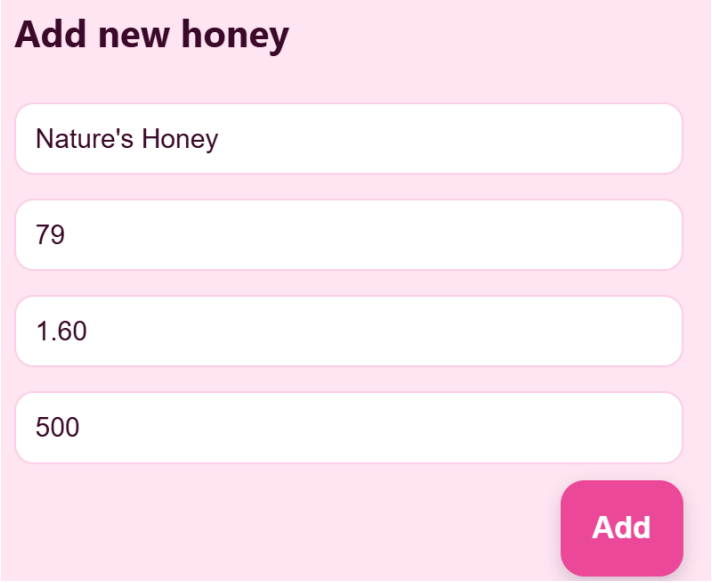
Sugar (%)

Price per bottle (£)

Mass per bottle (g)

Add

An example entry for a honey called “Nature’s Honey”, with 79 grams of sugar per 100g of honey, is £1.60 per bottle, and each bottle weighs 500 grams would be the following:



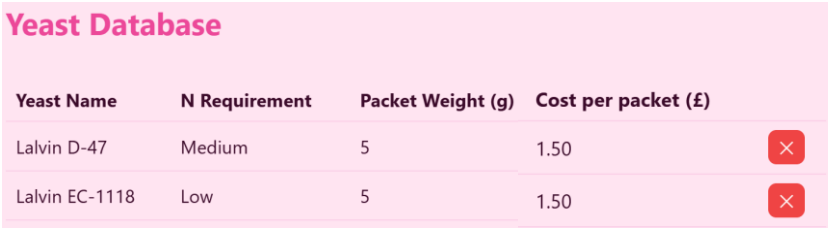
After pressing the “Add” button, the entry goes into the database where it becomes a dropdown option for the “Mead Recipe” page:

Honey name	Sugar (%)	Price per bottle (£)	Mass per bottle (g)	
Runny honey Lidl	79.7	1.59	400	×
Clear honey Lidl	79.7	1.69	500	×
Generic honey	80	1.49	300	×
Runny honey Rowse	80.8	5	720	×
Manuka honey Lidl	85.9	4.89	250	×
Nature's Honey	79	1.6	500	×

The red button on the right deletes the entry within that row, this may be used when that honey is no longer available for purchase or you no longer require that honey for any future brews. This is advised against due to that once the entry is deleted, it is not recoverable.

Yeast Database

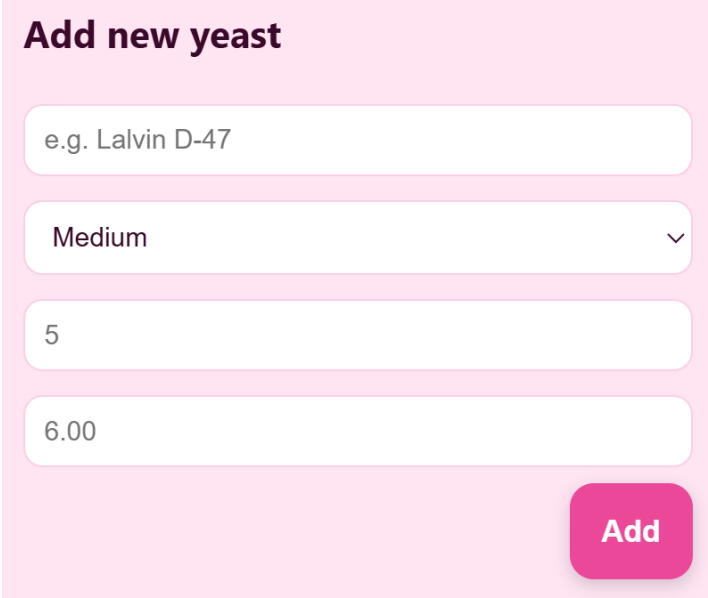
The next page is the yeast database. This holds all the required data for the app to carry out calculations in regard to the yeasts available. When the page is opened the following screen comes up:



It is built much like the “Honey Database” but with different data fields:

- N Requirement – this is the Nitrogen requirement of the yeast, this determines the amount of yeast nutrient the yeast needs during fermentation. This can be read from datasheets that the yeast producer provide online and can be found by googling the yeast and company name with “datasheet” in the search
- Packet weight – the weight of the yeast which is typically listed on the packet
- Cost per packet – like the honey, is based on where you buy your yeast from

And like the “Honey Database” there is also an entry field below the database for you to manually enter a new yeast:



And once again it works the exact same way as the “Honey Database” addition field does, this time for the “N Requirement” field it is a dropdown menu with 3 options, “Low”, “Medium”, and “High” depending on the specification sheets the yeast manufacturer provide.

pH Adjuster Database

This page again works in the same way as the previous two. Once you enter the page the following comes up:

Name	Type	H+ per mol	Molar mass (g/mol)	Notes	
Calcium carbonate ($CaCO_3$)	base	2	100.09	Raises pH	×
Citric acid	acid	3	192.12	Citrus/lemony	×
Malic acid	acid	2	134.09	Green apple acidity	×
Tartaric acid	acid	2	150.09	Grape-like acidity	×

The database is split into the following categories:

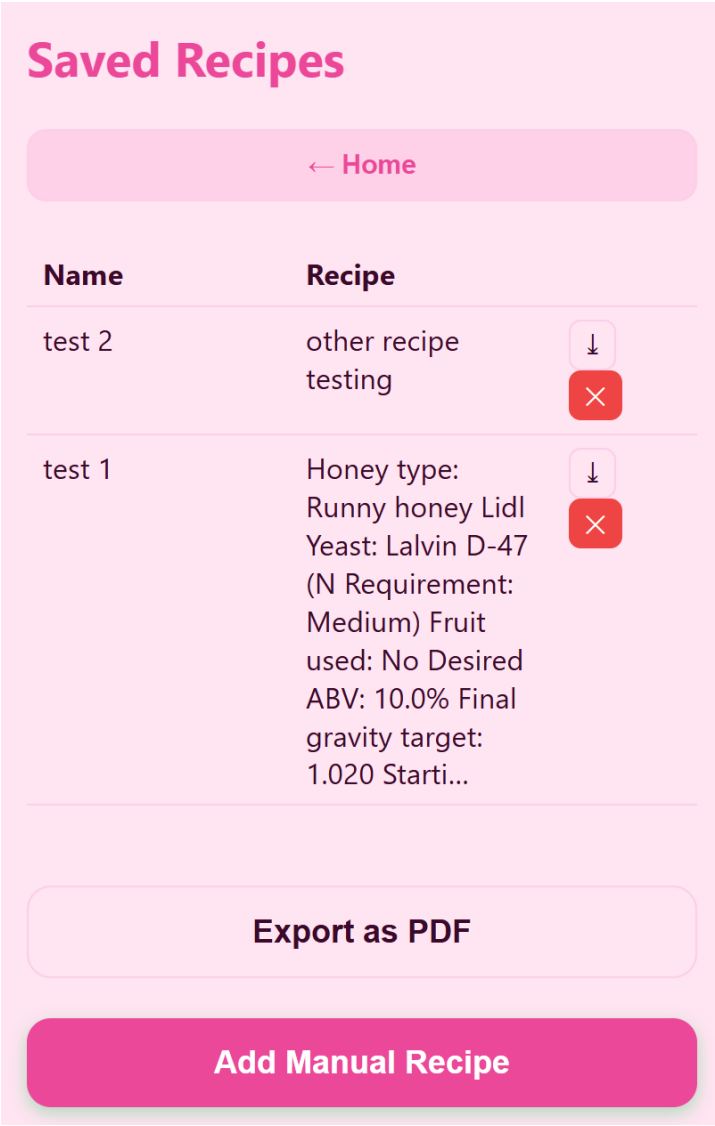
- Name – as usual this is just the name of the pH adjuster
- Type – either acid or base, this tells you if the pH adjuster increases pH or decreases the pH
- H+ per mol – this tells you the amount of [H^+] or [OH^-] ions that are produced per molecule of the pH adjuster, this can be found when googling the pH adjuster and googling the following "how many H+/OH- ions come from X molecule"
- Molar mass – this tells you the molecular weight of the pH adjuster, this can be found by googling the following "x molecular weight"
- Notes – this section is left open for whatever notes you can find on homebrewing/brewing pages about the pH adjuster, this is left for tasting notes that the pH adjuster will make to the brew so that way you can choose the adjuster based on the tasting notes to ensure a pleasant brew

And like the previous two databases there is an entry field at the bottom:

And once again it works the same way as the previous two databases did, with an acid/base dropdown with a note of how it adjusts the pH to ensure the correct one is selected.

Saved Recipes

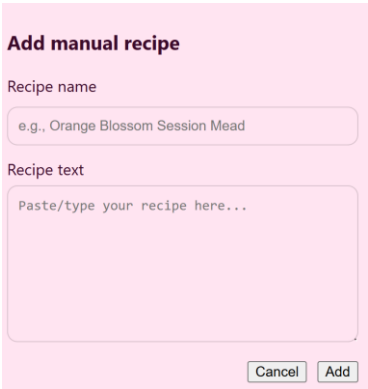
This page is where the "Save Recipe" button at the bottom of the "Mead Recipe" page sends the output to be stored for later use, once the page is opened the following comes up:



It only has two entries:

- Name – the name you would like to give the recipe
- Recipe – the actual recipe data produced

There is also a bonus button on the side above the delete button, this is the “Download” button and this brings up the recipe to a PDF print page so that way you can have a physical copy of the recipe so that way you can save it in a binder or folder.

This is also true for the “Export as PDF” button, however this button does it for all the recipes stored within this database.

The “Add Manual Recipe” button is like the entry fields in the previous databases, except this one brings up full text boxes: